Sami Hadouaj

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Summary

PhD Researcher focused on embedding probabilistic inference directly into database query execution engines. My research compiles structured probabilistic models into optimized inference algorithms that match the performance of hand-tuned implementations at scale. Built a full query execution engine in C++ on Apache Arrow with JIT compilation and parallel execution, validated on datasets exceeding 1 billion tokens. Two SIGMOD publications (2024, 2026).

Education

University of Michigan (Rackham Graduate School) — Dearborn, Michigan
Ph.D in Computer and Information Science, GPA: 3.96/4, Advised by Niccolò Meneghetti — Jan 2022 – 2026

Tunisia Polytechnic School — Tunis, Tunisia
Master of Science in Computer Science — Sep 2020 – Jun 2021

National Institute of Applied Science and Technology — Tunis, Tunisia
Bachelor in Software Engineering — Sep 2016 – Jun 2021

Experience

Applied Research Intern — May 2026 – Aug 2026
eBay — San Jose, California

- Built a hybrid vector and graph retrieval engine over eBay's knowledge graph to ground LLMs.

Research Experience

Graduate Research Assistant — Jan 2022 – Present
University of Michigan-Dearborn — Dearborn, Michigan

- Engineered a general-purpose probabilistic programming system in C++, integrating Apache Arrow for columnar in-memory data management, ClangJIT for just-in-time compilation of inference kernels, and OpenMP for multi-threaded parallel execution.
- Designed an automated compilation framework that synthesizes optimized Stochastic Variational Inference (SVI) algorithms from structured probabilistic model specifications, enabling scalable approximate Bayesian inference without manual algorithm engineering.
- Applied knowledge compilation techniques to eliminate redundant latent variables from probabilistic models, reducing the effective parameter space and accelerating convergence of variational inference.
- Benchmarked the framework on a Wikipedia corpus of over 1 billion tokens, demonstrating that the automatically synthesized inference algorithms achieve performance on par with specialized, hand-tuned implementations.

Undergraduate Research Assistant — May 2020 – May 2021
University of Michigan-Dearborn — Remote

- Engineered a recommendation engine that mapped bug reports to actionable code fixes, streamlining the debugging lifecycle for developers. Advised by Dr. Marouane Kessentini.

Publications & Technical Reports

- **Sami Hadouaj**, Ouael Ben Amara, Niccolò Meneghetti. "Variational Inference for De Finetti Logic" *Proceedings of the ACM on Management of Data* 4.3, Article 237 (June 2026), 30 pages. doi:10.1145/3802114 (**SIGMOD 2026**)
  [Paper] [Code]

- Ouael Ben Amara*, **Sami Hadouaj*** , Niccolò Meneghetti. "StarfishDB: a query execution engine for relational probabilistic programming" [1] *Proceedings of the ACM on Management of Data* 2.3 (2024): 1-31. doi:10.1145/3654988 (**SIGMOD 2024**)
  [Paper] [Code] [Poster] [Slides]

[1] Sami Hadouaj and Ouael Ben Amara are both first authors of this paper.

- **Sami Hadouaj**, Xiaojun Dong. "Reproducibility report for ACM SIGMOD 2024 paper: 'On efficient large sparse matrix chain multiplication' " *Reproducibility Reports of the 2024 International Conference on Management of Data.* 2025. doi:10.1145/3687998.3717039 (**SIGMOD 2024 Reproducibility**)
- Krause, Alexander, Georgiy Lebedev, **Sami Hadouaj**, Ouael Ben Amara, and Niccolò Meneghetti "Reproducibility report for ACM SIGMOD 2024 paper: 'StarfishDB: a query execution engine for relational probabilistic programming' " *Reproducibility Reports of the 2024 International Conference on Management of Data.* 2025. doi:10.1145/3687998.3717056 (**SIGMOD 2024 Reproducibility**)

Skills

Systems Engineering: C++ (primary), JIT compilation (ClangJIT), parallel computing (OpenMP), columnar data systems (Apache Arrow), build systems (CMake, Ninja), profiling & debugging (perf, gdb, Valgrind)

Languages & Tools: Python, SQL, Java, Datalog, Bash | Linux, Git, Docker

Research Areas: Uncertainty Quantification, Probabilistic Programming, Approximate Bayesian Inference (SVI, MCMC), Scalable Inference, Database Systems, Machine Learning, Generative Models, Topic Modeling, Natural Language Processing

Service

BIGDATA (2022, 2023, 2024, 2025): External reviewer

ICDM (2025): External reviewer

SIGMOD ARI (2024): Artifacts reviewer: Conducted independent code evaluation and experiment reproduction.

Volunteered with Accelerate4kids, developing educational materials to expand learning access for underprivileged children. (Sep - Dec 2024)

Teaching

Teaching Assistant, CIS 536: Text Mining & Information Retrieval (with Dr. Niccolò Meneghetti)	W2026
Lab Instructor, CIS 200: Software Engineering	F2025
Teaching Assistant, CIS 4951: Senior Design I and II (with Dr. Bruce Maxim)	Summer 2025
Teaching Assistant, CIS 556: Database Systems (with Dr. Niccolò Meneghetti)	F2023, W2024, F2024, W2025

Honors & Awards

Rackham Doctoral Intern Fellowship Program: Selected and funded by the University of Michigan Rackham Graduate School to conduct research during Summer 2024.

Talks & Presentations

SIGMOD 2024 Conference: *StarfishDB: A Query Execution Engine for Relational Probabilistic Programming*
Paper Presentation · Santiago, Chile · June 2024

Northeast Database Day 2026: *Stochastic Variational Inference for Relational Probabilistic Programming*
Poster Presentation · Boston, MA · January 2026

Northeast Database Day 2025: *Declarative Probabilistic Programming for Explainable AI*
Poster Presentation · Boston, MA · January 2025

SIGMOD 2024 Conference: *StarfishDB: A Query Execution Engine for Relational Probabilistic Programming*
Poster Presentation · Santiago, Chile · June 2024

Northeast Database Day 2024: *StarfishDB: A Query Execution Engine for Relational Probabilistic Programming*
Poster Presentation · Boston, MA · March 2024

Northeast Database Day 2023: *Query-driven Inference in Probabilistic Databases*
Poster Presentation · Boston, MA · March 2023